

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250277556

Kind Code

A1

Publication Date

September 04, 2025

Inventor(s)

Robertson; Jarrod S. et al.

Pipeline Passage Indicator

Abstract

Pipeline passage indicators for signaling the presence of a pig in a pipeline are disclosed. A pipeline passage indicator may house an upper magnet and a lower magnet in separate bores within the pipeline passage indicator. A flag attached to an upper end of the pipeline passage indicator is moveable from a first position to a second position in response to upward movement of the upper magnet. A labyrinth member defining a circuitous flow path is moveable from a first position to a second position in response to movement of a trigger from a first position to a second position. Movement of the labyrinth member will cause the lower magnet to move from its first position to its second position, which will cause the upper magnet to move from its first position to its second position and thereby move the flag from its first position to its second position.

Inventors: Robertson; Jarrod S. (Montgomery, TX), Sledd; Morgan M. (Conroe, TX), Segert; Andreas R. (Conroe, TX)

Applicant: Stark Solutions, LLC (Conroe, TX)

Family ID: 96880995

Assignee: Stark Solutions, LLC (Conroe, TN)

Appl. No.: 18/593977

Filed: March 03, 2024

Publication Classification

Int. Cl.: F16L55/48 (20060101)

U.S. Cl.:

CPC F16L55/48 (20130101);

Background/Summary

BACKGROUND OF THE INVENTIONS

1. Field of the Inventions

[0001] The present inventions generally pertain to oil and gas pipelines, and more particularly to devices for use in connection with pig cleaners and tools that are used to clean and inspect pipelines.

2. Description Of The Related Art

[0002] It is well known in the oil and gas industry that devices known as “pigs” are used to clean and inspect the interior of oil and gas pipelines. Pigs are placed inside the pipeline and then moved along the interior of the pipeline to clean the interior of the pipeline. A pig is inserted into a pipeline by first placing the pig into a chamber (sometimes referred to as a “launcher”) formed from a relatively short section of pipeline at an entry end of the pipeline. The chamber has an outer door and a valve, which is exposed to pressurized fluid within the pipeline to be cleaned. To insert the pig, the valve is closed to seal the chamber, and then the pig is inserted into the chamber between the outer door and the valve. Next, the valve is opened, which will cause the chamber to be filled with pressurized fluid from the pipeline. The pig is then moved into and along the pipeline to clean or inspect the pipeline. The pig will traverse the section of the pipeline to be cleaned or inspected, and eventually arrive at its destination, sometimes referred to as a “receiver” chamber where the pig is captured and removed from the pipeline.

[0003] It is important for the pig operator to know when the pig has passed key locations of the pipeline, such as when it arrives at the “receiver” chamber. It is well known within the oil and gas industry to provide what is known in the industry as a “pig signal” (referred to herein as a pipeline passage indicator) in the wall of the pipeline or launcher/receiver chamber to alert the pig operator when the pig has passed by a particular location. In simple terms, the pipeline passage indicator includes a trigger extending into the pipeline and within the path of the pig. When the pig passes the location of the pipeline passage indicator and hits the trigger, the trigger pushes a first or lower magnet up, which then pushes a second or upper magnet up, which causes a flag to move to an up position to indicate to the pig operator that the pig has passed that location. The top surface of the first or lower magnet and the bottom surface of the second or upper magnet have the same polarity. As such, when the lower magnet moves up into sufficiently close proximity to the upper magnet, the lower magnet will cause the upper magnet to move upwardly without touching it. The lower magnet is in a lower bore within the pipeline passage indicator, and the upper magnet is in an upper bore within the pipeline passage indicator. The upper and lower bores are separated from each other by an inner wall. As such, the two magnets are separated from each other by the inner wall, which has been designed to still allow the force of the lower magnet to push against the force of the upper magnet, when moved into close proximity of each other.

[0004] The lower bore is exposed to the chamber and to the pressurized fluids in the pipeline (and to fine debris entrained within the pressurized fluid) when that portion of the pipeline is pressurized. Because the lower magnet is in the lower bore, the lower magnet is likewise exposed to the pressurized fluid and debris within the pipeline. It is known within the industry that it is desirable to design pipeline passage indicators so as to minimize the amount of debris that can come into contact with and foul the lower magnet. If too much debris is allowed to cover the lower magnet, then the lower magnet will potentially no longer function for its purpose of pushing the upper magnet upwardly.

[0005] As will become apparent from the description and explanation set forth below, the present inventions address the above-described objective by providing new and improved pipeline passage indicators with new and improved designs to minimize the amount of debris that is allowed to

come into contact with the lower magnet.

SUMMARY OF THE INVENTIONS

[0006] In one aspect, the present inventions may include a pipeline passage indicator comprising: a body member including an upper internal bore and a lower internal bore; a flag attached to an upper end of the body member and moveable between a first position and a second position; an upper magnet disposed within the upper internal bore for movement between a first position and a second position; a lower magnet disposed within the lower internal bore for movement between a first position and a second position; a labyrinth member defining a circuitous flow path from a first end of the labyrinth member to a second end of the labyrinth member, and moveable between a first position and a second position; and a trigger disposed adjacent the first end of the labyrinth member and moveable between a first position and a second position. Another feature of this aspect of the present inventions may be that the labyrinth member includes a base member, a center disk, and a magnet holder, the lower magnet being at least partially positioned in a magnet bore in the magnet holder. Another feature of this aspect of the present inventions may be that the base member includes a support plate and a central stem, and the center disk includes a disk body member and a spacer extension extending from the disk body member, the spacer extension having a lower surface in contact with an upper surface of the support plate to define a flow space. Another feature of this aspect of the present inventions may be that the disk body member includes at least one flow aperture therethrough. Another feature of this aspect of the present inventions may be that the disk body member includes at least one pocket in a lower surface of the disk body member. Another feature of this aspect of the present inventions may be that the magnet holder includes a magnet body member, a lower surface of the magnet body member including at least one pocket disposed therein. Another feature of this aspect of the present inventions may be that the center disk includes a central passageway extending therethrough, and the magnet holder includes a central passageway extending therethrough, the central stem mating with and extending through the central passageway in the center disk and the central passageway in the magnet holder. Another feature of this aspect of the present inventions may be that the base member includes a support plate having an outer surface, the outer surface of the support plate including at least one groove, the center disk having a disk body member, the disk body member including at least one flow aperture therethrough, the at least one flow aperture being offset from the at least one groove.

[0007] In another aspect, the present inventions may include a pipeline passage indicator comprising: a body member including an upper internal bore, a lower internal bore, and an internal wall, the internal wall being disposed between the upper internal bore and the lower internal bore; a flag attached to an upper end of the body member and moveable between a first position and a second position; an upper magnet disposed within the upper internal bore for movement between a first position and a second position; a lower magnet disposed within the lower internal bore for movement between a first position and a second position; a base member disposed in the lower internal bore and including a central stem and a flow path; a center disk disposed in the lower internal bore, engaged with the central stem on the base member, and having a flow aperture offset from the flow path in the base member; a magnet holder disposed in the lower internal bore and engaged with the central stem on the base member, the lower magnet being at least partially positioned in a magnet bore in the magnet holder, the magnet holder including a flow path offset from the flow aperture in the center disk; and a trigger disposed adjacent the base member, and moveable between a first position and a second position. Another feature of this aspect of the present inventions may be that the base member further includes a support plate, and the center disk includes a disk body member and a spacer extension extending from the disk body member, the spacer extension having a lower surface in contact with an upper surface of the support plate to define a flow space. Another feature of this aspect of the present inventions may be that a lower surface of the center disk includes at least one pocket. Another feature of this aspect of the present inventions may be that the magnet holder includes a magnet body member, a lower surface of the

magnet body member including at least one pocket disposed therein. Another feature of this aspect of the present inventions may be that the center disk includes a central passageway extending therethrough, and the magnet holder includes a central passageway extending therethrough, the central stem on the base member mating with and extending through the central passageway in the center disk and the central passageway in the magnet holder. Another feature of this aspect of the present inventions may be that the base member includes a support plate having an outer surface, the outer surface of the support plate including at least one groove, the center disk having a disk body member, the disk body member including at least one flow aperture therethrough, the at least one flow aperture being offset from the at least one groove.

[0008] In another aspect, the present inventions may include a pipeline passage indicator comprising: a body member including an upper internal bore, a lower internal bore, and an internal wall, the internal wall being disposed between the upper internal bore and the lower internal bore; a flag attached to an upper end of the body member and moveable between a first position and a second position; an upper magnet disposed within the upper internal bore for movement between a first position and a second position; a lower magnet disposed within the lower internal bore for movement between a first position and a second position; a base member disposed in the lower internal bore and including a central stem and a support plate, the support plate having an outer surface, the outer surface on the support plate including at least one groove; a center disk disposed in the lower internal bore, engaged with the central stem on the base member, and having at least one flow aperture therethrough, the at least one flow aperture being offset from the at least one groove in the outer surface of the support plate; a magnet holder disposed in the lower internal bore and including a magnet body member and engaged with the central stem on the base member, the magnet body member having an outer surface including at least one groove, the at least one groove being offset from the at least one aperture in the center disk, the lower magnet being at least partially positioned in a magnet bore in the magnet body member; and a trigger disposed adjacent the base member, and moveable between a first position and a second position. Another feature of this aspect of the present inventions may be that the center disk includes a disk body member and a spacer extension extending from the disk body member, the spacer extension having a lower surface in contact with an upper surface of the support plate of the base member to define a flow space between the base member and the center disk. Another feature of this aspect of the present inventions may be that a lower surface of the center disk includes at least one pocket. Another feature of this aspect of the present inventions may be that a lower surface of the magnet body member includes at least one pocket disposed therein. Another feature of this aspect of the present inventions may be that the center disk includes a central passageway extending therethrough, and the magnet holder includes a central passageway extending therethrough, the central stem on the base member mating with and extending through the central passageway in the center disk and the central passageway in the magnet holder, and the base member, center disk, and magnet holder are connected to each other. Another feature of this aspect of the present inventions may be that the central stem on the base member has a square cross section, the central passageway of the center disk has square cross section configured to matingly engage with the central stem, and the central passageway of the magnet holder has a square cross section configured to matingly engage with the central stem.

[0009] Other features, aspects and advantages of the present inventions will become apparent from the following discussion and detailed description.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] FIG. 1 is a perspective view a specific embodiment of a pipeline passage indicator

constructed in accordance with the present inventions and shown with a flag indicator in a deployed position.

[0011] FIG. 2 is a side view of the pipeline passage indicator shown in FIG. 1 installed in the side wall of a section of pipeline with the flag indicator in a first or non-deployed position, and with an approaching pig in the pipeline and coming into initial contact with a trigger of the pipeline passage indicator, but before moving the trigger from a first or non-deflected position to a second or deflected position to cause the flag indicator to be moved into its second or deployed position.

[0012] FIG. 2A is a side view very similar to FIG. 2, except that the pig has now moved past the trigger and moved the trigger into its second or deflected position to cause the flag indicator to move into its second or deployed position.

[0013] FIG. 3 is a cross-sectional view of the pipeline passage indicator shown in FIGS. 1, 2 and 2A with the pipeline passage indicator in a first or non-deployed position.

[0014] FIG. 3A is an enlarged view of the indicated portion of FIG. 3.

[0015] FIG. 4 is a cross-sectional view similar to FIG. 3, but with the pipeline passage indicator now in a second or deployed position, with the trigger in its second or deflected position and the flag indicator in its second or deployed position.

[0016] FIG. 4A is an enlarged view of the indicated portion of FIG. 4.

[0017] FIG. 5 is a cross-sectional view taken along lines 5-5 of FIG. 3.

[0018] FIG. 6 is a cross-sectional view taken along lines 6-6 of FIG. 3.

[0019] FIG. 7 is a cross-sectional view taken along lines 7-7 of FIG. 3.

[0020] FIG. 8 is a cross-sectional view taken along lines 8-8 of FIG. 3.

[0021] FIG. 9 is a cross-sectional view taken along lines 9-9 of FIG. 3.

[0022] FIG. 10 is an exploded view showing a group of internal components of the pipeline passage indicator shown in FIGS. 3 and 4.

[0023] FIG. 11 is a perspective view of a specific embodiment of a unitary labyrinth member in accordance with one aspect of the present inventions.

[0024] While the inventions will be described in connection with the preferred embodiments, it will be understood that the scope of protection is not intended to limit the inventions to those embodiments. On the contrary, the scope of protection is intended to cover all alternatives, modifications, and equivalents as may be included within the spirit and scope of the inventions as defined by the appended claims.

DETAILED DESCRIPTION OF THE INVENTION

[0025] Referring to the drawings in detail, wherein like numerals denote identical elements throughout the several views, and referring initially to FIG. 1, there is shown a perspective view of a specific embodiment of a pipeline passage indicator **10** having a flag **12** shown in its second or deployed position. Referring now to FIG. 2, the pipeline passage indicator **10** is shown installed in a sidewall **14** of a pipeline. The pipeline passage indicator **10** includes a trigger **16** extending downwardly through an aperture **18** in the sidewall **14** and into an inner bore **20** of a pipeline **22**. In FIG. 2, the pipeline passage indicator **10** is shown in a first or non-deployed position, with the flag **12** in a first, non-deployed or horizontal position, and with the trigger **16** in its first or non-deflected position. FIG. 2 shows a pig **24** within the pipeline **22** at the point it makes initial contact with the trigger **16**. Referring now to FIG. 2A, the pig **24** is shown after it has made contact with the trigger **16**, which has caused the trigger **16** to move to its second or deflected position, which has caused the flag **12** to move to its second or deployed position.

[0026] Referring now to FIG. 3, a cross-sectional view of the pipeline passage indicator **10** is shown with the flag **12** in its first or non-deployed position and the trigger **16** in its first or non-deflected position. The pipeline passage indicator **10** may include an upper housing **26**, a lower housing **28**, and an inner body member **30**, each of which may be cylindrical in shape in a specific embodiment. The upper housing **26** may include an enclosed upper end **32** and an inner bore **34**. A lower surface of the inner bore **34** may be provided with a recess **36**. The lower housing **28** has an

inner bore **38**. An outer surface of an upper end of the lower housing **28** may include threads **40** disposed adjacent the recess **36**. The inner body member **30** may include an upper bore **42**, a lower bore **44**, and an inner wall **46**. The upper bore **42** extends downwardly from an upper end of the inner body member **30** and has a lower surface corresponding to an upper surface of the inner wall **46**. The lower bore **44** extends upwardly from a lower end of the inner body member **30** and has an upper surface corresponding to a lower surface of the inner wall **46**.

[0027] The upper end of the inner body member **30** may be disposed within the inner bore **34** of the upper housing **26**, and may be in contact with a lower surface of the enclosed upper end **32** of the upper housing **26**. An outer surface of the lower end of the inner body member **30** may include threads **48** adapted for threadable engagement with threads **50** disposed in the inner bore **38** of the lower housing **28**. A seal **52** (such as an annular seal) is positioned in a recess around an outer surface of the inner body member **30** and in sealing contact with the inner bore **38** of the lower housing **28** at a location above the mating threads **48** and **50**. In this manner, a sealed space is defined within the inner bore **38** of the lower housing **28** below the seal **52**. In this manner, pressurized fluid within the pipeline cannot move upwardly within the pipeline passage indicator **10** past the seal **52**. Some or all of the upper housing **26**, lower housing **28**, and inner body member **30** may be provided as a unitary structures as opposed to individual components.

[0028] Referring now to the lower end of the pipeline passage indicator **10**, the components contained within the pipeline passage indicator **10** within the sealed space below the seal **52** will now be discussed and described. More precisely, the components to be described now are situated within the inner bore **44** of the inner body member **30**. In a specific embodiment, it is noted that the lower end of the inner body member **30** may be provided with an extension member **54** that may be threadably attached to the inner body member **30** to effectively extend the inner bore **44** roughly down to a lower end of the lower housing **28**. The extension member **54** may also be integrally formed as part of the inner body member **30**. The inner bore **44** may have a lower enclosed end **56** (which may be part of the extension member **54**). The lower enclosed end **56** may include a trigger aperture **58** through which the trigger **16** may extend and be retained. The trigger **16** may include a head **60**, an angled transition section **62**, and a tip **64**. The trigger head **60** may include a plurality of flow recesses **61**, the purpose of which is to facilitate upward flow of pressurized fluid from the pipeline and past the trigger **16**. When in its first or non-deflected position as shown in FIG. 3, a lower surface of the trigger head **60** may be resting against and retained by an upper surface of the lower enclosed end **56** of the bore **44**, with the trigger tip **64** in a generally vertical orientation.

[0029] A labyrinth member, which may be an assembly of components or a unitary structure, is positioned above the trigger head **60** and within the lower bore **44** of the inner body member **30**. Referring now to FIGS. 3 and 10, in a specific embodiment, and starting at the bottom, the labyrinth member may include a base member **66**, a first center disk **68**, a second center disk **70**, a magnet retainer **72**, and a first or lower magnet **74**. In a specific embodiment, the labyrinth member may include only one of the center disks **68** and **70**. In another specific embodiment, the labyrinth member may include more than two center disks **68** and **70**. In a specific embodiment, the base member **66** may include a support plate **76**. In another specific embodiment, the base member **66** may include a central stem **78** extending upwardly from a center of the support plate **76**. The support plate **76** may have a circular cross section. The central stem **78** may have a square or rectangular cross section. A central bore **80** extends upwardly through the center of the support plate **76** and through the center of the central stem **78**. A lower section of the central bore **80** may be threaded, as can be seen, for example, in FIG. 3. An outer surface of the support plate **76** may include one or more grooves **82** to act as fluid flow paths, as will be discussed further hereinbelow. In a specific embodiment, the support plate **78** may include three grooves **82**, and the grooves **82** may be in the shape of angular notches. In another specific embodiment, the support plate **76** may include one or more flow apertures through the body of the support plate **76** to act as flow paths instead of including grooves on the outside of the support plate **76**.

[0030] Moving upwardly now to the next component, the first center disk **68** may include a disk body member **84** and a spacer extension **86** extending downwardly from the disk body member **84**. In a specific embodiment, each of the disk body member **84** and the spacer extension **86** may have a circular cross section. The disk body member **84** and the spacer extension **86** may include a central passageway **88**, which may extend from a lower surface of the spacer extension **86** to an upper surface of the disk body member **84**. The central passageway **88** may be adapted for mating engagement with the central stem **78** of the base member **66** (i.e., in a specific embodiment, the central passageway **88** may have a square or rectangular cross section). The disk body member **84** may also include one or more flow apertures **90** extending from a lower surface of the disk body member **84** to an upper surface of the disk body member **84**. In a specific embodiment, the disk body member **84** may include three flow apertures **90**. In a specific embodiment, the flow apertures **90** may be offset from or not aligned with the grooves **82** (or flow apertures) in the base member **66**. In a specific embodiment, the flow apertures **90** may be positioned approximately equidistant between the grooves **82** (or flow apertures) in the base member **66**. In a specific embodiment, the location of the flow apertures **90** relative to the grooves **82** may be fixed by virtue of the geometric relationship between the shape of the central stem **78** on the base member **66** and the shape of the central passageway **88** through the first center disk **68**. In the specific embodiment shown, those shapes are square or rectangular. But the scope of the present inventions is not limited to any particular shape, but instead covers any shape or configuration that will prevent relative rotation of the base member **66** and the first center disk **68** when assembled. In another specific embodiment, a fastener (e.g., a screw, bolt, pin, etc.) may be used instead of the central stem **78** to secure the various components together (i.e., the base member **66** may be included without a central stem **78**). The purpose of offsetting the flow apertures **90** in the first center disk **68** relative to the grooves **82** (or flow apertures) in the base member **66** is to create a circuitous flow path of pressurized fluid coming from the pipeline through the labyrinth member, as will be further discussed below.

[0031] Still referring to the first center disk **68**, in a specific embodiment, the lower surface of the disk body member **84** may include a plurality of debris pockets **92** extending partially up into the lower surface of the disk body member **84**. The purpose of the debris pockets **92** is to capture and retain debris contained within the pressurized fluid flowing upwardly from the pipeline into the labyrinth member, to thereby prevent it from reaching the first/lower magnet **74**. In a specific embodiment, the pockets **92** may be circular in shape, but the present inventions are not limited to any particular shape, as any shape is intended to be covered by the present inventions. Furthermore, instead of pockets **92**, other alterations to the structure of the lower surface of the disk body member **84** may be provided for the purpose of trapping solid contaminants and debris, such as machining a spiral into the lower surface, adding a texture to the lower surface, or attaching a fabric material to the lower surface, such as a felt material. As best shown in FIG. 3, when the labyrinth components are assembled (if the labyrinth member is not a unitary structure), the lower surface of the spacer extension **86** rests on top of the upper surface of the base member **66**, and creates a flow space **93**, such as an annular space, around the spacer extension **86** and between the upper surface of the base member **66** and the lower surface of the disk body member **84**. The pressurized fluid coming from the pipeline up into the labyrinth member will flow within the flow space **93** as it is diverted on its circuitous path from the grooves **82** (or flow apertures) in the base member **66** to the flow apertures **90** in the first center disk **68**.

[0032] Moving up to the next component, the second center disk **70** may be structurally very similar to the first center disk **68**, in that the second center disk **70** may include a disk body member **94**, a spacer extension **96**, a central passageway **98**, one or more flow apertures **100**, and a plurality of debris pockets **102**. The difference is that the flow apertures **100** in the second center disk **70** are offset from the flow apertures **90** in the first center disk **68**. This is done for the same purpose as explained above, i.e., to extend the circuitous flow path of pressurized fluid coming from the pipeline. In a specific embodiment, where pockets **102** are included, this may also be done

for the purpose of directing fluid flow over the pockets **102** so the pockets **102** can catch and retain debris contained within the pressurized fluid and prevent it from reaching the first/lower magnet **74**. Again, as mentioned above, in a specific embodiment, a pipeline passage indicator may be provided without the second center disk **70**. Also, as mentioned above, in a specific embodiment, the pipeline passage indicator may be provided with more than two center disks.

[0033] Moving up to the next component, the magnet retainer **72** may be structurally similar to the first center disk **68** and the second center disk **70**, in that the magnet retainer **72** may include a magnet body member **104**, a spacer extension **106**, a central passageway **108**, and a plurality of debris pockets **110**. Instead of flow apertures **90/100**, the magnet body member **104** may include grooves **112** such as the grooves **82** in the base member **66**. However, all that is needed for each of the base member **66**, the first center disk **68**, the second center disk **70**, and the magnet retainer **72** to have some fluid flow passageway (e.g., an outer groove, a flow aperture, or some other flow path to create a circuitous flow path from a first or lower end of the labyrinth member or assembly to a second or upper end of the labyrinth member or assembly). In a specific embodiment, the magnet body member **104** may include at least one flow aperture instead of grooves **112**. Another difference is that the magnet retainer **72** includes a magnet bore **114** (shown in FIG. **3**) for holding the first/lower magnet **74**. In a specific embodiment, a fastener **116** (e.g., screw, bolt, pin, etc.) may extend through a central bore in the first/lower magnet **74** and into the central bore **80** of the central stem **78** of the base member **66**, and threadably engage with the threaded portion of the central bore **80** of the central stem **78**, to hold the components of the labyrinth assembly together. The bolt is not necessary if the labyrinth member is a unitary structure. A lower spring **118** is provided to impart a downward force on the labyrinth member. In a specific embodiment, the spring **118** may be partially disposed around the first/lower magnet **74** and between an upper surface of the lower bore **44** and an upper surface of the magnet retainer **72**.

[0034] As explained above, the labyrinth member may be provided as an assembly of separate components that are assembled into a single component, or it may be formed as a unitary structure. In either case, the labyrinth member defines a circuitous flow path from a first or lower end of the labyrinth member to a second or upper end of the labyrinth member. A specific embodiment of a unitary labyrinth member **138** is shown in FIG. **11**. Referring now to FIG. **11**, if the labyrinth member **138** is a unitary structure, in a specific embodiment, it may include a base section **140**, at least one center disk section (such as a first center section **142** or a second center section **144**), and a magnet holder section **146**, each of which may be separated by flow spaces **148**, **150**, **152** (which could be between surfaces of those sections that are in contact with one another). In a specific embodiment, a circuitous flow path may extend from the first or lower end of the labyrinth member **138** (such as a lower surface of the base section **140**) through a flow path in the base member (such as one or more grooves **154**), into the flow space **148** between the base section **140** and the first center disk section **142**, through a flow aperture **156** in the first center disk section **142** that is offset from the flow path **154** in the base member, into the flow space **150** between the first center disk section **142** and the second center disk section **144**, through a flow aperture **158** that is offset from the flow aperture **156** in the first center disk section **142**, into the flow space **152** between the second center disk section **144** and the magnet holder section **146**, and through at least one flow path through the magnet holder section (such as a groove **160**) that is offset from the flow aperture **158** in the second center disk section **144**, and into a space above the second or upper end of the labyrinth member (e.g., an upper surface of the magnet holder section **146**). In a specific embodiment, at least one of lower surfaces of the center disk sections **142** and **144** and the magnet holder section **146** may include at least one debris pocket **162**, **164**, **166** for collecting and housing debris as pressurized fluid flows through the circuitous path. In a specific embodiment, the labyrinth member **138** may include multiple center disk sections, with flow apertures offset from adjacent flow apertures or flow paths. In a specific embodiment, the labyrinth member **138** may not include any center disk sections, in which case the circuitous flow path is defined by a flow path

through or around the base section into an offset flow path through or around the magnet holder section **146**.

[0035] Focusing now on the upper portion of the pipeline passage indicator **10**, a second or upper magnet **120** is located in the upper bore **42** of the inner body member **30**. An upper magnet extension **122** may be secured to an upper surface of the upper magnet **120**, such as with a bolt **124**. A spring **123** may be disposed around the upper magnet extension **122** and between a lower surface of the enclosed upper end **32** and an upper end of the upper magnet **120** to urge the upper magnet **120** downwardly within the upper bore **42**. An upper end of the upper magnet extension **122** extends through an aperture **126** in the enclosed upper end **32** of the upper housing **26**. An upper surface of the upper magnet extension **122** contacts a rocker arm **128**, which is rotatably mounted above the enclosed upper end **32** of the upper housing **26** on a pin **130**. The rocker arm **128** has a first or non-deployed position as shown in FIG. 3, which corresponds to the first, down, or non-deployed position of the flag **12**. The rocker arm **128** has a second or deployed position as shown in FIG. 4, which corresponds to the second, up, or deployed position of the flag **12**.

[0036] The operation of the specific embodiment of the pipeline passage indicator **10** shown in the Figures and discussed above will now be described. Referring to FIG. 2, the pig **24** is shown approaching and making initial contact with the tip **64** of the trigger **16**. At this point, the trigger **16** is in its first or non-deflected position, and the flag **12** is in its first/down/non-deployed position; and, as shown in FIG. 3, the lower magnet **74** and the upper magnet **120** are in their respective first or lower positions. It is also noted that, at this point, the interior space of the pipeline passage indicator **10** that is below the seal **52** is now occupied by and exposed to pressurized fluid from the pipeline. Referring now to FIG. 2A, the pig **24** has now moved past the trigger **16** and deflected the trigger **16** to its second or deflected position, as shown in FIG. 4. With reference to FIG. 4, when the trigger **16** is moved into its second or deflected position, an upper right corner **132** of the trigger head **60** will come into contact with and upwardly push a lower surface of the base member **66**. This will push the lower magnet **74** upwardly toward the inner wall **46**, and into sufficiently close proximity to the upper magnet **120** to push the upper magnet upwardly. This is due to the like polarities of the lower magnet **74** and the upper magnet **120**. This will push the upper magnet extension **122** upwardly to cause the rocker arm **128** to rotate upwardly around pin **130**. This will disengage the rocker arm **128** from a flag assembly that includes a spring **134** that will push the flag **12** into its second, up or deployed position, as shown in FIGS. 2A and 4.

[0037] From the above discussion and descriptions of the present inventions, it can be seen that the present inventions provide improved pipeline passage indicators that provide improved isolation of the lower magnet **74** from contaminants and debris contained within the pressurized fluid in the pipeline. This is accomplished by providing a circuitous flow path through a labyrinth or maze so that debris and contaminants in the pressurized fluid to which the lower magnet **74** is exposed is minimized. In this manner, the degree to which the lower magnet **74** is fouled by or covered with debris is minimized, which thereby extends the effective life of the lower magnet **74** to perform its function of pushing the upper magnet **120** up to release the flag **12**, as discussed above. This can further be enhanced by positioning one or more debris pockets **92/102/110** along the circuitous pass to filter out and house debris within the pressurized fluid.

[0038] It is to be understood that the inventions disclosed herein are not limited to the exact details of construction, operation, exact materials or embodiments shown and described. Although specific embodiments of the inventions have been described, various modifications, alterations, alternative constructions, and equivalents are also encompassed within the scope of the inventions. Although the present inventions may have been described using a particular series of steps, it should be apparent to those skilled in the art that the scope of the present inventions is not limited to the described series of steps. The specification and drawings are, accordingly, to be regarded in an illustrative rather than a restrictive sense. It will be evident that additions, subtractions, deletions, and other modifications and changes may be made thereunto without departing from the broader

spirit and scope of the inventions as set forth in the claims set forth below. Accordingly, the inventions are therefore to be limited only by the scope of the appended claims. None of the claim language should be interpreted pursuant to 35 U.S.C. 112(f) unless the word “means” is recited in any of the claim language, and then only with respect to any recited “means” limitation.

Claims

1. A pipeline passage indicator comprising: a body member including an upper internal bore and a lower internal bore; a flag attached to an upper end of the body member and moveable between a first position and a second position; an upper magnet disposed within the upper internal bore for movement between a first position and a second position; a lower magnet disposed within the lower internal bore for movement between a first position and a second position; a labyrinth member defining a circuitous flow path from a first end of the labyrinth member to a second end of the labyrinth member, and moveable between a first position and a second position; and a trigger disposed adjacent the first end of the labyrinth member and moveable between a first position and a second position.
2. The pipeline passage indicator of claim 1, wherein the labyrinth member includes a base member, a center disk, and a magnet holder, the lower magnet being at least partially positioned in a magnet bore in the magnet holder.
3. The pipeline passage indicator of claim 2, wherein the base member includes a support plate and a central stem, and the center disk includes a disk body member and a spacer extension extending from the disk body member, the spacer extension having a lower surface in contact with an upper surface of the support plate to define a flow space.
4. The pipeline passage indicator of claim 3, wherein the disk body member includes at least one flow aperture therethrough.
5. The pipeline passage indicator of claim 3, wherein the disk body member includes at least one pocket in a lower surface of the disk body member.
6. The pipeline passage indicator of claim 2, wherein the magnet holder includes a magnet body member, a lower surface of the magnet body member including at least one pocket disposed therein.
7. The pipeline passage indicator of claim 2, wherein the center disk includes a central passageway extending therethrough, and the magnet holder includes a central passageway extending therethrough, the central stem mating with and extending through the central passageway in the center disk and the central passageway in the magnet holder.
8. The pipeline passage indicator of claim 2, wherein the base member includes a support plate having an outer surface, the outer surface of the support plate including at least one groove, the center disk having a disk body member, the disk body member including at least one flow aperture therethrough, the at least one flow aperture being offset from the at least one groove.
9. A pipeline passage indicator comprising: a body member including an upper internal bore, a lower internal bore, and an internal wall, the internal wall being disposed between the upper internal bore and the lower internal bore; a flag attached to an upper end of the body member and moveable between a first position and a second position; an upper magnet disposed within the upper internal bore for movement between a first position and a second position; a lower magnet disposed within the lower internal bore for movement between a first position and a second position; a base member disposed in the lower internal bore and including a central stem and a flow path; a center disk disposed in the lower internal bore, engaged with the central stem on the base member, and having a flow aperture offset from the flow path in the base member; a magnet holder disposed in the lower internal bore and engaged with the central stem on the base member, the lower magnet being at least partially positioned in a magnet bore in the magnet holder, the magnet holder including a flow path offset from the flow aperture in the center disk; and a trigger disposed

adjacent the base member, and moveable between a first position and a second position.

10. The pipeline passage indicator of claim 9, wherein the base member further includes a support plate, and the center disk includes a disk body member and a spacer extension extending from the disk body member, the spacer extension having a lower surface in contact with an upper surface of the support plate to define a flow space.

11. The pipeline passage indicator of claim 9, wherein a lower surface of the center disk includes at least one pocket.

12. The pipeline passage indicator of claim 9, wherein the magnet holder includes a magnet body member, a lower surface of the magnet body member including at least one pocket disposed therein.

13. The pipeline passage indicator of claim 9, wherein the center disk includes a central passageway extending therethrough, and the magnet holder includes a central passageway extending therethrough, the central stem on the base member mating with and extending through the central passageway in the center disk and the central passageway in the magnet holder.

14. The pipeline passage indicator of claim 9, wherein the base member includes a support plate having an outer surface, the outer surface of the support plate including at least one groove, the center disk having a disk body member, the disk body member including at least one flow aperture therethrough, the at least one flow aperture being offset from the at least one groove.

15. A pipeline passage indicator comprising: a body member including an upper internal bore, a lower internal bore, and an internal wall, the internal wall being disposed between the upper internal bore and the lower internal bore; a flag attached to an upper end of the body member and moveable between a first position and a second position; an upper magnet disposed within the upper internal bore for movement between a first position and a second position; a lower magnet disposed within the lower internal bore for movement between a first position and a second position; a base member disposed in the lower internal bore and including a central stem and a support plate, the support plate having an outer surface, the outer surface on the support plate including at least one groove; a center disk disposed in the lower internal bore, engaged with the central stem on the base member, and having at least one flow aperture therethrough, the at least one flow aperture being offset from the at least one groove in the outer surface of the support plate; a magnet holder disposed in the lower internal bore and including a magnet body member and engaged with the central stem on the base member, the magnet body member having an outer surface including at least one groove, the at least one groove being offset from the at least one aperture in the center disk, the lower magnet being at least partially positioned in a magnet bore in the magnet body member; and a trigger disposed adjacent the base member, and moveable between a first position and a second position.

16. The pipeline passage indicator of claim 15, wherein the center disk includes a disk body member and a spacer extension extending from the disk body member, the spacer extension having a lower surface in contact with an upper surface of the support plate of the base member to define a flow space between the base member and the center disk.

17. The pipeline passage indicator of claim 15, wherein a lower surface of the center disk includes at least one pocket.

18. The pipeline passage indicator of claim 15, wherein a lower surface of the magnet body member includes at least one pocket disposed therein.

19. The pipeline passage indicator of claim 15, wherein the center disk includes a central passageway extending therethrough, and the magnet holder includes a central passageway extending therethrough, the central stem on the base member mating with and extending through the central passageway in the center disk and the central passageway in the magnet holder, and the base member, center disk, and magnet holder are connected to each other.

20. The pipeline passage indicator of claim 19, wherein the central stem on the base member has a square cross section, the central passageway of the center disk has square cross section configured

to matingly engage with the central stem, and the central passageway of the magnet holder has a square cross section configured to matingly engage with the central stem.
